Meredith LeFevre

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EDUCATION

University of Iowa — **August 2023 – (Anticipated) May 2027**

Bachelor of Science in Physics and Astronomy, Minors in Mathematics, Geoscience, and Dance
GPA: 3.91 cumulative

AWARDS

Girl Scout Gold Award — **October 2022**

Dean's List, University of Iowa — **Fall 2023 – Present**

Independent Creative Research by Undergraduates (ICRU) Fellowship, University of Iowa — **Fall 2026 – Spring 2027**

Phi Beta Kappa, Alpha of Iowa Chapter, University of Iowa — **Spring 2026 – Present**

National Scholars Award — **August 2023 – May 2027**

RESEARCH PROJECTS

Red Supergiants in M31: Investigating Binary Candidates in Cluster and Field Environments — **Summer 2026 – Present**

Undergraduate Researcher – Advisor: Dr. Cliff Johnson, Northwestern University (CIERA REU)

- Identified red supergiants (RSGs) in M31 cluster and field populations using the Panchromatic Hubble Andromeda Treasury and the visually identified Andromeda Project (AP) star cluster catalog
- Developed cluster-specific color-magnitude diagram selection criteria to identify RSGs within individual AP star clusters
- Identified candidate binary systems through ultraviolet excess consistent with hot B/OB companions
- Currently investigating completeness effects and comparing observed RSG populations and candidate binary fractions with stellar population models
- Manuscript in preparation, with findings expected to be reported in Spring 2027: https://meredith-lefevre.github.io/files/CIERA_REU_Research_Summary_Draft.pdf

Classifying Galactic Center Sources Through X-ray Time-Domain Analysis — **Spring 2026 – Present**

Undergraduate Researcher – Advisor: Dr. Dustin Swarm, University of Iowa

- Investigate Galactic Center X-ray sources using Chandra X-ray Observatory data to distinguish magnetic and non-magnetic cataclysmic variables
- Perform time-domain analysis of X-ray light curves for ~34 sources to identify periodic variability
- Use spin and orbital periodicity to improve source classification and refine white dwarf mass estimates

The Iowa Idea at Great Heights - Iceland Solar Eclipse High-Altitude Ballooning Project — **Spring 2026 – Summer 2026**

Data and Instrumentation Lead

- Led instrumentation strategy for a high-altitude balloon mission in Iceland (July 2026), integrating a magnetometer to measure geomagnetic field strength within the auroral oval
- Developed calibration procedures, sampling strategy, and onboard data logging systems for geomagnetic measurements
- Participated in flight operations and post-launch analysis following a cutdown system failure that prevented payload recovery and retrieval of flight data

Tully-Fisher Relation Analysis and Regression Methods **Fall 2024 - Spring 2025**

Undergraduate Researcher – Advisor: Dr. Hai Fu, University of Iowa

- Compared HyperFit and LMFIT for modeling the Tully-Fisher Relation, evaluating statistical assumptions, treatment of uncertainties, and fitting approaches
- Applied these methods to forward, inverse, and unified Tully-Fisher models, investigating the effects of treating galaxy inclination as a fixed parameter
- Examined how linear regression and galaxy inclination assumptions influence interpretation of the Tully-Fisher Relation

PROFESSIONAL EXPERIENCE

Undergraduate Teaching Assistant – Clear Skies, University of Iowa **Spring 2026 - Present**

- Assisted with instruction and student engagement for General Education astronomy courses
- Supported astronomy outreach events, public observing nights, and nighttime sky demonstrations
- Provided hands-on support for astronomy laboratory and observational activities

PSAPS (Prairie Section of the American Physical Society) Annual Meeting Organizing Committee, University of Iowa **Spring 2026 - Fall 2026**

- Organized scientific and undergraduate programming for the PSAPS Annual Meeting (October 22–24, 2026), including sessions in astronomy, plasma, particle, and materials physics
- Coordinated outreach events, career panels, website development, invited speaker outreach, and conference logistics, including securing discounted hotel rates
- Contacted local SPS chapters to promote the conference and helped lead an undergraduate SPS welcome event

Outreach Coordinator: Society of Physics Students, University of Iowa **Fall 2025 - Present**

- Organize SPS social and department-wide events, including faculty colloquiums, Trivia Night, and an REU/Graduate School Application Workshop
- Coordinate volunteers and promote student participation in public observing and physics and astronomy outreach opportunities
- Manage SPS communication through weekly meetings, email, social media, and event coordination to strengthen member engagement

Undergraduate STEM Ambassador, Iowa Space Grant Consortium **Fall 2025 - Present**

- One of 11 undergraduate students selected statewide to represent ISGC in promoting NASA's mission through STEM outreach and community engagement
- Represent the Iowa Space Grant Consortium through STEM outreach, professional development, and collaboration with students, educators, and community partners across Iowa

POSTERS & PRESENTATIONS

"Red Supergiants in M31: Investigating Binary Candidates in Cluster and Field Environments" **January 2027**

249th American Astronomical Society Meeting, Salt Lake City, UT

"Classifying Galactic Center Sources Through X-ray Time-Domain Analysis" **November 2026**

Fall Undergraduate Research Festival, University of Iowa

"Red Supergiants in M31: Investigating Binary Candidates in Cluster and Field Environments" **October 2026**

Prairie Section of the APS Annual Meeting, University of Iowa

"Red Supergiants in M31: Investigating Binary Candidates in Cluster and Field Environments" October 2026
Society of Physics Students Undergraduate Colloquium, University of Iowa

"Red Supergiants in Andromeda: Uncovering Their Hidden Companions" October 2026
Public Observing Night Presentation Public Talk, University of Iowa

"Red Supergiants in M31: Investigating Binary Candidates in Cluster and Field Environments" August 2026
Adler Astronomy Conversations, Adler Planetarium

"Red Supergiants in M31: Investigating Binary Candidates in Cluster and Field Environments" August 2026
CIERA REU Poster Presentation, Northwestern University

"Dead Stars That Still Shine: X-rays from the Galactic Center" June 2026
Public Observing Night Presentation Public Talk, University of Iowa

TECHNICAL TRAINING

Programming Languages & Tools:

Python (NumPy, Matplotlib, Jupyter Notebook), LaTeX, CIAO (Chandra X-ray analysis), Unix/Linux Command Line

Discovery Science Center Summer School for Matter at Extreme Conditions, Center for Matter at Atomic Pressures, University of Rochester June 2026
Attendee

- Participated in an intensive program on extreme states of matter in laboratory and astrophysical systems
- Applied Python to investigate topics in astrophysics, planetary science, and quantum materials
- Engaged with researchers and gained exposure to computational tools in high-energy/density physics

Coding the Cosmos: Introduction to Gravitational Waves, Missouri University of Science and Technology May 2024
Attendee

- Analyzed gravitational wave data using Python for scientific computation and visualization
- Explored compact object mergers, gravitational wave detection, and analysis methods used by LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA

Van Allen Observatory Training, University of Iowa April 2024

- Trained on the University of Iowa's 17-inch f/6.8 Cassegrain telescope mounted on a PlaneWave L-500 direct-drive Alt-Az mount
- Gained experience with SBIG 6303E CCD imaging system and LRGB filter set for astrophotography and observation
- Operated 6-inch Celestron telescope with internal alignment software and 4-inch Orion telescopes for educational observing and public outreach

SCIENCE OUTREACH EXPERIENCE

Department of Physics and Astronomy Outreach Committee, University of Iowa Spring 2026 – Present

- Coordinate outreach logistics and volunteer efforts for observing nights and outreach events
- Help improve departmental outreach communication, scheduling, and event planning

Physics Demonstration Volunteer – Hawkeyes on Science & Hawkeyes in Space, University of Iowa — **Fall 2025 – Present**

Volunteer

- Present interactive physics and astronomy demonstrations to K–12 students and community audiences
- Explain core concepts including electromagnetism, waves, optics, and space science using hands-on experiments
- Adapt scientific explanations for diverse age groups to promote STEM accessibility and engagement

Project HOPE (Healthcare Occupations Preparation and Exploration), University of Iowa — **May 2026**

Volunteer

- Helped facilitate robotics sessions for middle school students through hands-on soldering activities
- Guided groups of ~10 students in basic electronics, tool safety, and STEM problem-solving

Helio & Space Physics Outreach Committee, University of Iowa — **Fall 2025**

- Led solar observing activities, provided educational materials, and engaged the public using 3D-printed models of NASA's TRACERS mission instruments
- Facilitated an interactive exhibit on the history of the University of Iowa's space missions, inviting alumni and visitors to share their experiences and connections to space science

Cub Scout Astronomy Outreach, University of Iowa — **October 2025**

Volunteer

- Led the "Cub Scout Adventure" astronomy elective, introducing first-grade students to basic observational astronomy concepts
- Guided scouts in observing the night sky and identifying constellations
- Demonstrated the use of telescopes and binoculars for sky observation

Partial Solar Eclipse Day, University of Iowa — **April 2024**

Volunteer

- Set up and managed telescopes and outreach equipment at the University of Iowa Sciences Library for the April 8th Solar Eclipse
- Answered visitor's questions about eclipse safety, telescope use, and basic solar astronomy

Public Observing Nights, University of Iowa — **Fall 2024 - Present**

Volunteer

- Set up and take down of telescopes, signage, and outreach materials
- Assist with public engagement and explain basic telescope anatomy and astronomical concepts

CONFERENCES & WORKSHOPS

APS Global Physics Summit, Denver Colorado — **March 2026**

Attendee

APS CU*iP (Conference for Undergraduate Women and Gender Minorities in Physics), Denver, Colorado — **March 2026**

Michigan Technological University — **January 2025**

Attendee